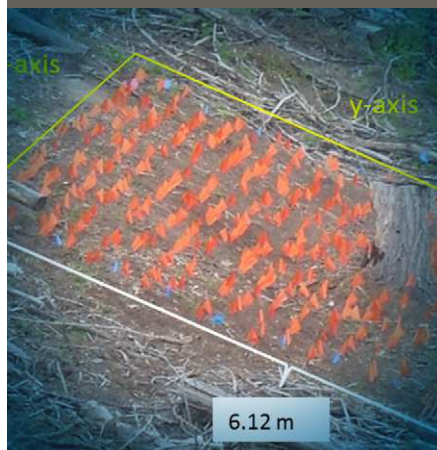


Special Section: Frontiers
of Hydrogeology in Vadose
Zone Research

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We investigated the use of pseudo-three-dimensional electrical resistivity tomography (ERT) as a field method for characterizing soil–pipe spatial distribution and connectivity on a forested hillslope, with an interest in detecting the formation of soil pipes from tree decay and root combustion. We conclude that it is not practical to rely on homogeneity assumptions commonly used to detect anomalies characterized by higher resistivities compared to the surrounding soil. The use of ERT surveys to detect large macropores may be more suitable to areas of soil where the pathways are not occurring very near and within portions of a restrictive layer.

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Characterizing Soil–Pipe Networks with Pseudo-Three-Dimensional Resistivity Tomography on Forested Hillslopes with Restrictive Horizons

Soil pipes are important hillslope–hydrology features affecting mechanistic processes such as subsurface drainage, streamflow, erosion, hillslope stability, and discharge response to rain fall and snow melt. However, the limited capacity to characterize these pathways in situ remains a major limitation for process-based understanding and incorporation into coupled hydrologic models. The goal of this study was to investigate the use of pseudo-three-dimensional electrical resistivity tomography (ERT) as a field method for characterizing soil–pipe spatial distribution and connectivity on a forested hillslope with a specific interest in detecting the formation of soil pipes from root decay and combustion. We conducted detailed ERT surveys on three 6.12- by 6.12-m plots before and after a prescribed burn located on a 2-ha clearcut of coniferous forest underlain by a fragipan. Excavation following the ERT measurements showed that soil pipes were mostly empty with 86% containing <25% decaying root matter. The mean diameter of pipes was 4.3 cm with a mean length of 1.6 m. Such empty pipes would resolve as elongated high-electrical resistivity features. However, a high resistivity layer corresponding to the Bw and BE soil horizons overlaying a fragipan was observed on all plots with the majority of soil pipes larger than 1 cm diameter occurring within these horizons. The lack of observable contrast between pipes and the high resistivity layer as well as reduced data coverage within the vicinity of large soil pipes may have limited our ability to effectively characterize soil pipes occurring within these hillslopes. We conclude that the use of ERT surveys to detect large macropores pathways may be more suitable to settings where soil pipes are not occurring within high restrictive layers.

Abbreviations: ERT, electrical resistivity tomography; GPR, ground penetrating radar; UIEF, University of Idaho Experimental Forest

The hydrological importance of soil pipes (Fig. 1) in forested hillslopes, lack of in situ information of their spatial connectivity and distribution, and lack of suitable tools for mapping and monitoring of soil pipes motivated this investigation of ERT as a possible technique for studying the distribution and connectivity of these important hydrological pathways.

Soil pipes comprise preferential pathways and interconnected macropores with cm-scale diameters and varying lengths that may form through, or be enhanced by, biological activity (Nieber et al., 2006), freeze-thaw processes (Carey and Woo, 2002), internal erosion (Wilson, 2011), or the decay and combustion of roots (Leslie, 2012). Past research has focused primarily on soil pipe hydrologic effects in peat soils (Bryan and Jones, 1997) and temperate forests (Uchida et al., 2001) largely with a view on hydrologic responses of preferential pathways (e.g., Parlange et al., 1989, Anderson et al., 2009a). In these systems, soil pipes have been identified as important hydrologic features affecting subsurface drainage (Tromp-van Meerveld and McDonnell, 2006), streamflow (Uchida et al., 1999), erosion (Wilson, 2011), hillslope stability (Uchida et al., 2001), discharge response to snow melt (Roberge and Plamondon, 1987), threshold driven hillslope runoff from storm rainfall (Uchida et al., 2005), and the release of soil carbon (Billett et al., 2012).

As one of the dominant processes in hillslope hydrology, the ability to incorporate these preferential flow pathways into water resource modeling is essential for predicting watershed responses to changes in land use and related phenomena including climate change (Sidle et al., 2001; Pilling and Jones, 2002; Weiler and McDonnell, 2007). However, as



Fig. 1. Soil pipes formed from decaying tree roots at plot 1 within the first 50 cm of a coarse-silty, mixed, superactive, frigid Vitrandic Fragixeralfs profile. Rulers display decimeter scale, used for picture scale only.

stated in a recent review paper by Lin (2010), without new methods of identifying preferential flow pathways in situ, we are at a technological bottle neck restricting the advancement of our understanding and ability to model preferential flow. The problem is further compounded by transient and dynamic changes in soil pipe networks due to biotic, erosional, and other processes (Sidle et al., 2001; Uchida et al., 2001; Wilson, 2011).

Most studies mapping macropore preferential flow pathways have relied on a combination of dyes, tracers, and excavation (e.g., Noguchi et al., 1997, 1999; Anderson et al., 2009b), with the latter being labor intensive and time consuming, limiting the extent of soil pipe characterization to the pedon scale. Furthermore, excavation destroys the soil structure, eliminating the possibility of monitoring changes in soil pipe distribution and connectivity over time. While the characterization of soil pipe distribution and connectivity with nondestructive or minimally invasive techniques would be preferable for examining the relationships between land use, climate, hydrological processes, and soil pipe evolution, few studies have examined indirect techniques for investigating soil pipe distribution, connectivity, and morphology. Perhaps the most detailed study on soil pipe morphology was undertaken in Hokkaido, Japan, by Terajima et al. (2000) in which a fiber optic scope was used to examine the connectivity and geometry of soil pipes up to a meter in length. However, the investigation of soil pipes beyond a meter

in length with the fiber optic scope was limited by soil pipe geometry. In another study, Holden (2004) used a geophysical method, ground penetrating radar (GPR), with a sodium chloride tracer for detecting soil pipes in blanket peat and a sandy loam. The authors' concluded that the method was limited by the need for preexisting knowledge of the soil pipe location for injection of the saline tracer for detection.

Ground penetrating radar and another geophysical method, ERT, have been used to detect living root systems (al Hagrey, 2007; Zenone et al., 2008) and estimate root biomass (Butnor et al., 2001; Amato et al., 2008). Some studies have even demonstrated the ability to detect individual roots and estimate root diameters under optimal laboratory conditions using GPR (Barton and Montagu 2004) and ERT (Zanetti et al., 2011). However, under field conditions, signals originating from individual roots are often indistinguishable from noise. This is particularly true when gravel and pebbles are present (Zenone et al., 2008). Hirano et al. (2009) showed that roots with high water content were detectable if greater than 19 mm in diameter but that GPR was not able to detect roots with less than 20% water content as would be expected for decaying roots or empty root channels under unsaturated soil conditions. Much of the problem stems from the challenges of collecting data on cm-scale grids; even when using a millimeter-accuracy positioning system for the three-dimensional GPR survey

by Yokota et al. (2011), only roots with diameters greater than 5 cm could be detected reliably.

In this study, we assess the utility of using ERT to characterize soil–pipe networks using the example of decayed tree roots, which form soil pipes networks in coniferous forests. These decayed roots may further be enhanced as well as initiated (opened up to the surface) with forest fires (Leslie, 2012). Clearcut timber harvesting techniques leave behind uniformly aged stumps and roots, which coupled with subsequent controlled and uncontrolled fires, enhance the development of soil pipes from root decay and combustion. In addition, Beven and Germann (1982) suggested that depressions occurring from former stump locations could funnel water into macropore networks from decaying tree root systems. Such initiated soil pipes forming within forested watersheds may be particularly important in combination with water-restricting soil horizons or bedrock that form seasonally-perched water tables. Such water-restricting layers are common in northern Idaho in the form of fragipans and similar argillic horizons (McDaniel et al., 2001, 2008). Soil pipe formation in forest soils from the decay of tree roots has been previously described in several studies (e.g., Aubertin, 1971; Noguchi et al., 1997, 1999; Tsuboyama et al., 1994). Yet, detailed investigations of soil–pipe morphology and distribution within coniferous hillslopes of the United States have not yet occurred despite their prevalence in some forest ecosystems (Leslie, 2012). The ERT method (e.g., Samouelian, 2005) is a potentially promising tool to image soil pipe distribution and connectivity in situ. The ERT surveys are minimally invasive and resolve changes in soil–electrical resistivity associated with changes in soil texture, structure, and water content (e.g., Friedman, 2005). Electrical resistivity tomography has been used to study live roots (Zenone et al., 2008) and preferential flow (Oberdörster et al., 2010), detecting these features and processes as high electrical conductivity anomalies. In contrast, empty or partially filled soil pipes are expected to resolve as spatially elongated high resistivity anomalies when compared to the background soil. Such anomalies have been mapped successfully in the past including empty sink holes (Panno et al., 1994), buried cellars (Nyari and Kanli, 2007), tomb cavities (Elwaseif and Slater, 2010), badger dens (Butler and Roper, 1994), and desiccation cracks (Samouelian et al., 2003), but until now, no attempt to examine ERT as a field method for mapping soil pipe networks has been made.

Materials and Methods

Study Area

Our field site was located on a 2-ha clearcut enclosing a portion of a small drainage located in the Flat Creek Unit of the University of Idaho Experimental Forest (UIEF), approximately 11 km north of Troy, Idaho at 46° 50 29.01 N lat, 116° 46 29.65 W long and at 940 m above sea level (Fig. 2). The UIEF ranges in daily mean temperature from 0°C to 17°C from winter to summer, respectively, and receives approximately 690 mm annual precipitation

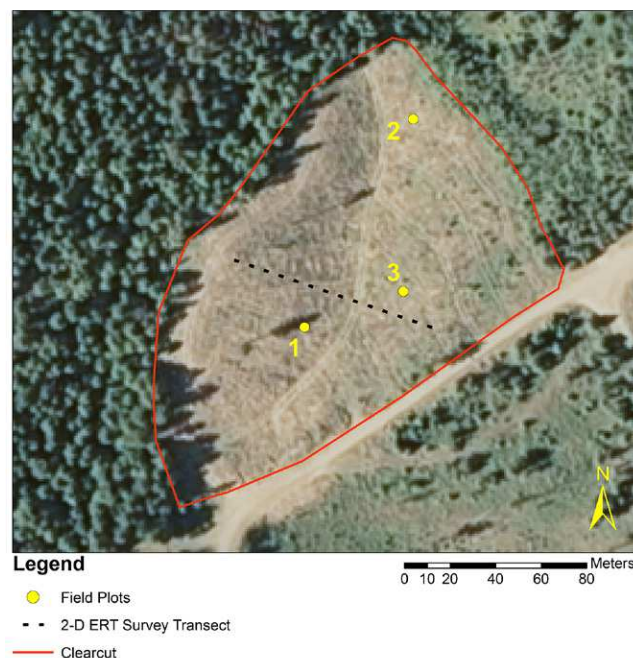


Fig. 2. Location of electrical resistivity tomography (ERT) field plots 1, 2, 3 on the clearcut study area at the Flat Creek Unit of the University of Idaho Experimental Forest. Also shown is the location of the two-dimensional ERT transect (black dashed line).

(Sandoval et al., 2007). Over 60% of annual precipitation in the region occurs as light rain and snow from November to April (Brooks et al., 2004).

Timber species before harvesting consisted of a diverse conifer forest composed of grand fir [*Abies grandis* (Douglas ex D. Don) Lindl.], western redcedar (*Thuja plicata* Donn ex D. Don), Douglas-fir (*Pseudotsuga menziesii* (Mirb.) Franco), ponderosa pine (*Pinus ponderosa* P. Lawson & C. Lawson), and western larch (*Larix occidentalis* Nutt.). Timber was last harvested the summer of 2009 using a cut-to-length harvester-forwarder method. Fir, larch, and pine were previously harvested in the area from 1927 to 1929. The study site underwent a prescribed burn using a strip-firing technique on 14 Oct. 2010.

Before the burn, we selected three 6.12- by 6.12-m plots for detailed ERT surveys (Fig. 2). Plot locations were chosen to maximize the potential for detection of soil pipes. Each plot was therefore centered on an individual tree stump to minimize the superposition of root architectures from multiple trees as well as to maximize the potential for tree root combustion (Fig. 3). Plots were on 33, 27, and 31% slopes, for plots 1, 2, and 3, respectively. Plot 1 was on a northeast facing slope and plots 2 and 3 were on northwest facing slopes.

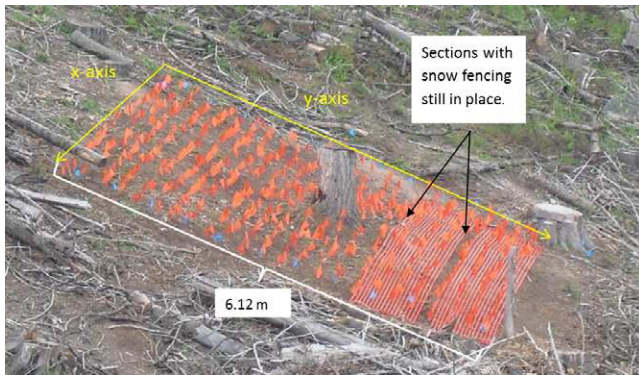


Fig. 3. Plot 1 before the prescribed burn. Flags are placeholders for stainless steel electrodes replaced before measurements along linear transects in a roll-along fashion in both the x - and y -direction. Marked snow fencing was used as a template to accomplish accurate and precise electrode placement along an orthogonal grid in the face of microtopography and debris.

Soil Descriptions

The soils are mapped as the Santa Series, and are ashy silt loams on 3 to 35% slopes classified as coarse-silty, mixed, superactive, frigid Vitrandic Fragixeralfs (Soil Survey Staff, 2004; Soil Survey Division, 2003). A soil profile description was obtained from an excavation one meter to the side of plot 1 at the same hillslope position, while soil profile descriptions for plots 2 and 3 were obtained during the excavation of the plots. Table 1 provides selected soil morphological and physical characteristics from the field plots. The O horizon is composed of slightly to moderately decomposed pine needles, bark, and grasses, and ranges in depth from 0 to 4 cm. A thin layer of relatively unweathered volcanic ash (C horizon) from the 1980 Mt. St. Helens eruption ranges in thickness from 0.5 to 1 cm. The O and C horizons are not uniformly present across the plots, possibly having been disturbed during the timber harvesting operations. The A horizon for these soils are thin and very similar to the Bw horizon with the exception of many more very fine and fine roots from grasses and forbs and breaking to

Table 1. Summary of soil morphological and physical characteristics.

Horizon	Depth	Dry color	Texture	Structure†	Rupture resistance
	cm				
Plot 1					
O	0–4	NA	NA	NA	NA
C	4–4.5	2.5YR 7/1	NA	NA	NA
A	4.5–10	very pale brown (10YR 7/3)	silt loam	2 m sbk parting to 1 f g	slightly hard
Bw	10–28	very pale brown (10YR 7/3)	silt loam	2 m sbk	slightly hard
BE	28–46	pale brown (10YR 6/3)	silt loam	1 f sbk	slightly hard
E	46–65	very pale brown (10YR 7/3)	silt loam	massive	slightly hard
Btxb1	65–90	very pale brown (10YR 7/3)	silt loam	2 c sbk	very hard
Btxb2	90–102+	light yellowish brown (10YR 6/4)	silty clay loam	2 c abk	very hard
Plot 2					
O	0–2.5	NA	NA	NA	NA
C	2.5–3.5	light gray (2.5YR 7/1)	NA	NA	NA
A	3.5–10	brown (7.5YR 4/3)	silt loam	1 m sbk parting to 1 f g	slightly hard
Bw1	10–23	brown (7.5YR 4/3)	silt loam	1 m sbk	slightly hard
Bw2	23–37	brown (7.5YR 5/3)	silt loam	1 m sbk	slightly hard
BE	37–46	brown (10YR 5/3)	silt loam	1 f sbk	slightly hard
E	46–55	very pale brown (10YR 7/3)	silt loam	massive	slightly hard
Btxb	55–60+	yellowish brown (10YR 6/4)	silt loam	1 f sbk	very hard
Plot 3					
O	0–4	NA	NA	NA	NA
C	4–5	light gray (2.5YR 7/1)	NA	NA	NA
A	5–8	brown (7.5YR 4/3)	silt loam	1 m sbk parting to 1 f g	slightly hard
Bw	8–30	pale brown (10YR 6/3)	silt loam	2 m sbk	slightly hard
BE	30–45	pale brown (10YR 6/3)	silt loam	1 f sbk	slightly hard
E	45–51	very pale brown (10YR 7/3)	silt loam	massive	slightly hard
Btxb	51–62 +	yellowish brown (10YR 6/4)	silt loam	1 f sbk	very hard

† 1 = weak, 2 = moderate, f = fine, m = medium, c = coarse, g = granular, sbk = subangular blocky, abk = angular blocky, NA = not applicable.

granular structure. Each soil profile exhibited one to two Bw horizons between 8 and 37 cm in depth, followed by a transitional BE horizon extending in depth up to 46 cm. The E horizon ranged from 45 to 65 cm in depth. All soil profiles exhibited a water-restricting Btxb (fragipan) horizon beginning between 51 and 65 cm in depth and running roughly parallel to the slope of the soil surface. The fragipan horizons at the clearcut are believed to have originated from a Pleistocene paleosol and are common throughout the Palouse region of eastern Washington and northern Idaho (McDaniel et al., 2001, 2008). The coarse to very coarse prismatic structure typically observed with Btxb horizons in these soils was not observed at plots 2 and 3, however the Btxb horizon was restrictive to roots, dense, brittle, with very hard rupture resistance.

Electrical Resistivity Data Acquisition

A pseudo-three-dimensional resistivity survey was conducted at each of the plots before and after the prescribed burn over a 2-d period. Preburn plot resistivity surveys were conducted beginning on 15 and 22 July 2010 and 8 Aug. 2010 and postburn plot resistivity surveys on 21 and 27 June 2011 and 7 July 2011 for plots 1, 2, and 3, respectively. Electrical resistance readings were obtained using a multichannel ERT system GeoTom MK-RES/IP/SP (GEOLOG2000, Starnberg, Germany) along 25 parallel lines (two-dimensional transects) in both the x and y directions using two parallel lines of 25 electrodes at a time for later three-dimensional reconstruction. To help ensure electrode spacing accuracy, snow fencing was used to mark electrode locations at 0.255 m square intervals before measurements (Fig. 3).

We used a combination of Wenner- β and dipole-dipole electrode arrays: specifically a Wenner- β with an a -spacing from 2 to 7, and a dipole-dipole array with an a spacing equal to 1 and $n = 1 - 7$. The dipole-dipole array was selected due to its good horizontal resolution, which is desirable for detecting cavities (Dahlin and Zhou, 2004; Loke, 2011). An additional dipole-dipole array with a $2a$ spacing for $n = 4 - 7$ was used to help increase the signal strength for the larger n factors (Loke, 2011). The Wenner- β array is a version of the dipole-dipole array where spacing between all electrodes remains the same (Loke, 2011). The dipole-dipole array is also advantageous for pseudo-three-dimensional ERT surveys in that the array has a relatively quick data acquisition time compared to other arrays and should be able to tolerate a larger spacing between the two-dimensional survey lines given the array's longer sensitivity pattern (Loke, 2011).

Sub-centimeter-accuracy topographical measurements of the plots were collected to facilitate calculation of geometry factors and apparent resistivity data used during the inversion process. The topography of each of the three plots was obtained by measuring the relative change in elevation at each electrode position from a common reference point using a water level attached to a pressure transducer (SMS tensiometer, Soil Measurement Systems, Tucson, AZ).

To validate ERT reconstructions in imaging soil pipes, plots were excavated by hand to a minimum depth of 0.50 m in 0.255 m increments parallel to the x axis along the 25 profiles documenting soil pipe location, connectivity, and geometries. Excavations of soil plots were conducted from 11 Aug.–3 Oct. 2011. For more detailed methods and results, we refer the reader to Leslie (2012); here we provide a short summary.

Excavated soil pipes showed that 96% of all soil pipes showed evidence of forming from decaying tree roots, with more than 86% containing <25% decaying root matter. The mean diameter of soil pipes were 3.8, 5.0, and 4.1 cm for plots 1, 2, and 3, respectively. Median tortuosity of all soil pipes was 1.23 with a mean path length of 1.6 m. Soil pipe occurrence increased with depth up to between 30 and 40 cm and with a center frequency at a depth of 32 cm corresponding to the beginning of the BE horizon.

Additionally, a longer two-dimensional resistivity transect traversing the clearcut was obtained on 15 Sept. 2010 before the prescribed burn, using the same electrode arrays as in the plot surveys with the electrode spacing increased to 1 m (Fig. 2 and 6).

ERT Data Analysis

Inverse solution reconstruction of the resistivity distribution from the apparent resistivity data was conducted using the Boundless Electrical Resistivity Tomography (BERT) ver. 1.2.4 software package, based on the finite element-forward modeling and inversion method described in Rücker et al. (2006) and Günther et al. (2006).

Robust Inversion and Regularization Parameter

Before inversion, we identified outliers using an iterative implementation of the Grubbs Test (Grubbs, 1969) with $\alpha = 0.05$ on log-transformed data, and removed outliers only when they were not clustered spatially. Inversions were performed using a robust inversion optimization method (L_1 -norm) motivated by the need to resolve soil pipes as sharp resistivity contrasts. The robust inversion method has been found to be less sensitive to noise in the data and produce fewer artifacts in the inversion resistivity images than the smooth inversion optimization method (L_2 -norm) (Dahlin and Zhou, 2004; Zhou and Dahlin, 2003). Loke et al. (2003) concluded that the L_1 -norm was more suitable than the L_2 -norm for areas containing sharp resistivity boundaries. The regularization parameter λ was chosen by the operator to produce the model with the best fit between the data and model roughness. While the choice of λ is somewhat arbitrary and relies on the experience of the operator, there is typically a range of acceptable λ values and the setting of λ by the operator is a standard procedure for geophysical inversion (Udphuay et al., 2011). A summary of important mesh and inversion options for BERT are given in Table 2.

Table 2. Inversion and mesh options for inversion in BERT.

	Inversion options		Mesh options			
	λ	Robust data (L1)	Para3/ two-dimensional quality†	Para depth	Prim3/ two-dimensional quality†	PrimP2mesh (quadratic mesh)
Three-dimensional plot tomograms	15	1 = yes	1.3	3	1.2	1 = yes
Two-dimensional plot tomograms	10	1 = yes	34	3	33.4	1 = yes
Two-dimensional Clearcut tomogram	L-curve‡	1 = yes	34	auto	33.4	1 = yes

† Three- and two-dimensional quality parameters on different scales; see <http://www.resistivity.net/download/bert-tutorial.pdf>.

‡ See Günther et al. (2006) for a description of the L-curve regulation.

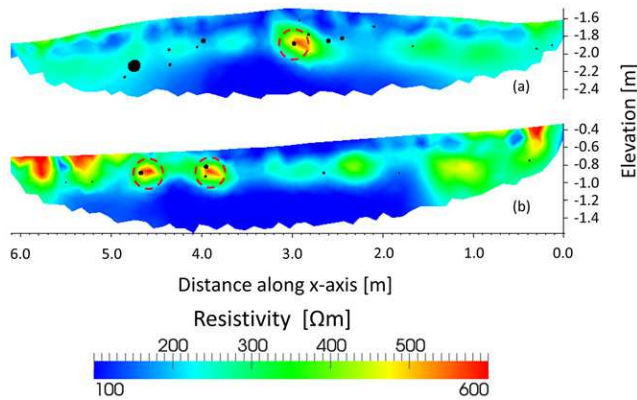


Fig. 4. Two-dimensional resistivity tomograms from individual post-burn x axis resistivity transects; (a): plot 3 profile 21 ($y = 4.84$ m); (b): plot 3 profile 6 ($y = 1.21$ m); black closed circles indicate actual soil pipe diameter and location; open dashed red circles surround high resistivity anomalies where soil pipes were within a 0.10 m radius of the anomaly.

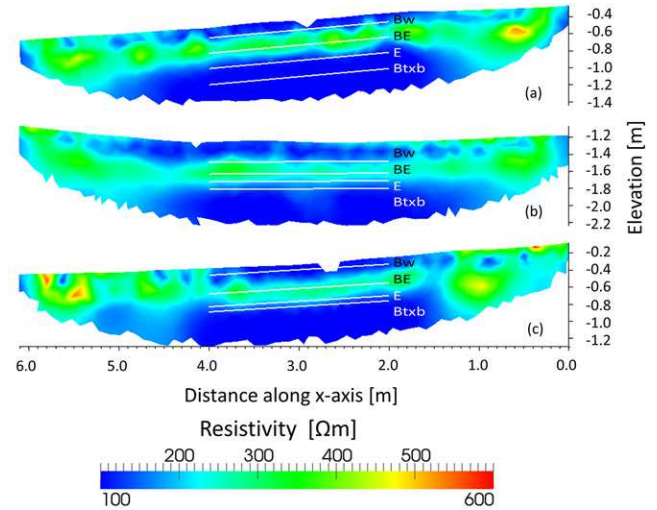


Fig. 5. Two-dimensional resistivity tomograms from individual post-burn x axis transects with selected soil horizons; (a): plot 1, (b): plot 2, (c): plot 3.

Visualization and Spatial Analysis

Two-dimensional and three-dimensional inversion tomograms were visualized using ParaView (2013). Two-dimensional and three-dimensional tomograms are composed of an unstructured triangular mesh and unstructured three-dimensional volumetric mesh, respectively. Visualizations of two-dimensional tomograms were limited to a minimum data coverage value of 1.5 except for the two-dimensional transect across the clearcut, which was limited to a minimum coverage value of 1 due to an increase in electrode spacing and decreased resistivity model resolution. The coverage value, the summation of the absolute sensitivities, represents the uncertainty associated with the resistivity model, with higher values having greater certainty.

Results and Discussion

A first examination of the resistivity data for soil pipe locations was undertaken through the inversion of apparent resistivity measurements on individual transects parallel to the x axis, which comprised half of the three-dimensional resistivity data set, resulting in two-dimensional resistivity tomograms (Fig. 4). Tomograms

were visually inspected for high resistivity anomalies spatially corresponding with documented soil-pipe locations from excavations and morphological studies indicated as black circles in Fig. 4. Anomalies occurring close to soil pipe openings were observed in two of the 75 post-burn tomograms, as exhibited within the red circles of Fig. 4. The spatial locations of the soil pipe openings match fairly well—within 0.10 m of the centers of the high resistivity areas; however, these three corresponding locations from two transects do not provide enough information to determine whether the high resistivity anomalies are truly occurring from the soil pipes, or are related to other changes in soil properties.

The vertical center of the approximately-parallel to the soil surface high-resistivity layer corresponds with the transition from the Bw to the BE soil horizons at each plot, which begins at a depth of 28, 37, and 30 cm for plots 1, 2, and 3, respectively (Fig. 5). This high resistivity layer is characteristic for this site as indicated in Fig. 6, which shows the continuity across an extensive portion of this two-dimensional tomogram transecting the 2-ha clearcut.

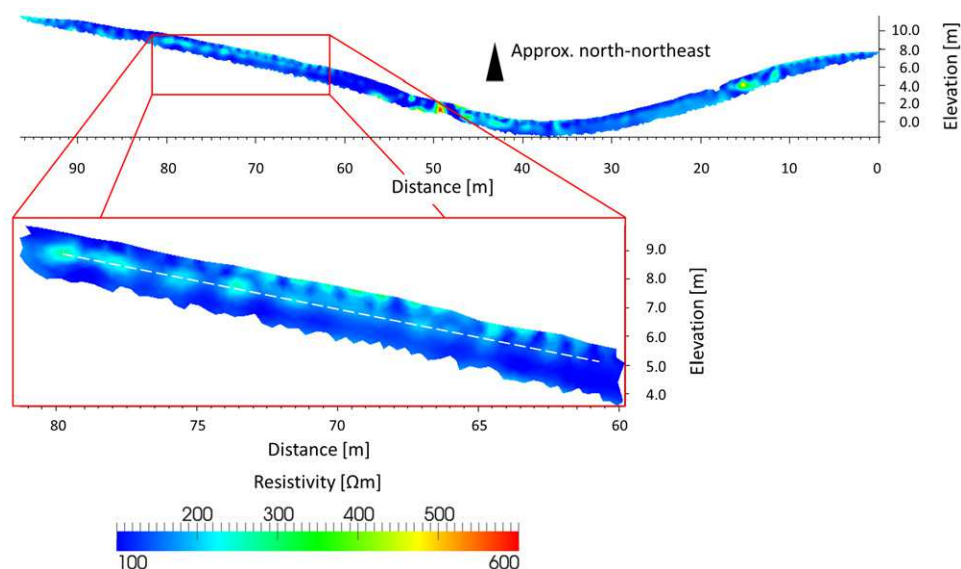


Fig. 6. Two-dimensional resistivity tomogram from the two-dimensional transect across the clearcut (see Fig. 2) with a close up exhibiting the high resistivity layer near the surface along the white dashed line. The high resistivity layer observed from the two-dimensional tomogram of the clearcut is not as well defined due to the decreased resolution from the increase in electrode spacing to 1 m compared to the 0.225 m electrode spacing used for the plot resistivity surveys.

The increase in resistivity with the start of the BE horizon may be partially attributable to an increase in soil density with resulting reduction in soil water content where the resistivity response is more sensitive to the reduction in water content compared to the increase in bulk density (Laloy et al., 2011). Reuter et al. (1998) reported mean dry-soil bulk densities ranging from 1.49 to 1.62 Mg m^{-3} for the Bw horizon and mean bulk densities $>1.6 \text{ Mg m}^{-3}$ for the E, Btb, and Btxb horizons in similar Fragixeralfs and Argixerolls of the Palouse region in Northern Idaho. The high bulk densities typical of these soils, and weak to massive structure of the Bw and BE soil horizons (see Table 1), suggest that low water contents contributed to the high resistivity layer corresponding to these horizons. On the other hand, the low resistivity area with around $100 \text{ } \Omega\text{m}$ beneath the high resistivity layer in Fig. 4–6 is likely an artifact due to the overlaying high resistivity layer. This is a common occurrence in resistivity tomograms when anomalies are close to the surface (Dahlin and Loke, 1997).

The high resistivity layer was also observable in three-dimensional electrical-resistivity tomograms from datasets obtained before and after the prescribed burn. Figure 7 shows segmented post-burn images with collocated soil pipes and tree stumps. Bandpass filtering of the images using the *threshold* filter in ParaView was used to examine the three-dimensional tomograms for elongated high resistivity anomalies corresponding to connected soil pipes through the selection of ranges of resistivity for voxels. The *threshold* filter was also used to search for soil pipes through the selection of cells within discrete resistivity ranges starting at the low end of the resistivity distribution ($25 \text{ } \Omega\text{m}$) and moving towards the high end ($1000 \text{ } \Omega\text{m}$) using discrete intervals of $5 \text{ } \Omega\text{m}$. No high resistivity anomalies corresponding to the documented soil-pipe locations were visually identifiable.

The lack of high resistivity anomalies corresponding to collocated soil-pipe crosssections is partially explained by Fig. 8, which shows frequency distributions of resistivity as a function of depth from three-dimensional tomograms of each plot examined at approximately 300,000 point locations evenly distributed in 0.06 m intervals along the surface topography and in 0.02 m intervals along the z -axis, from -0.02 to -0.80 m . Resistivities extracted from voxels with collocated soil pipes (irrespective of size) were not significantly different ($p > 0.25$) for any of the pre- or post-burn plots from soil background resistivities using a two-sided t test with $\alpha = 0.05$ after log-transforming for normality. In addition, soil pipe voxels were not significantly different between pre- and post-burn images indicating that there was no corresponding increase in resistivity associated with the generation of soil pipes due to combustion. Leslie (2012) showed that while soil pipes close to the stump were enhanced by fire, that soil pipes at larger distances from the stump were attributed to form primarily (96%) through root-organic matter decay. These soil pipes were primarily free of organic matter, debris, or soil, but had thin sections of bark or organic stains lining the macropores (Leslie, 2012). It is therefore not surprising that pre- and post-burn comparisons of resistivity images did not resolve the formation of soil pipes at large. Furthermore, the presence of the high resistivity layer may substantially impair our ability to resolve anomalies such as soil pipes, which show similar high resistivities because of lack in contrast. In this layer, resistivity values approaching $316 \text{ } \Omega\text{m}$ ($2.5 \log_{10}[\Omega\text{m}]$ in Fig. 8) occur with greater frequencies at depths up to about 0.40 m , which corresponds to the Bw/BE soil horizons.

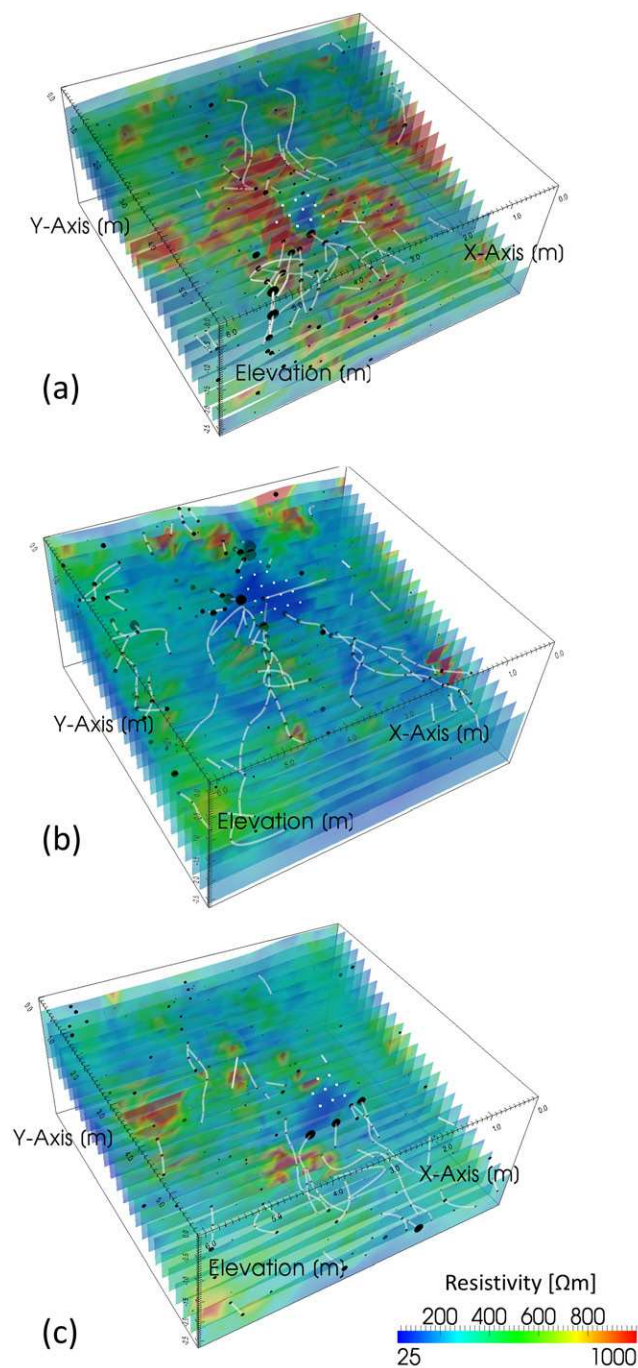


Fig. 7. Post-burn three-dimensional resistivity tomograms: plot 1 (a), plot 2 (b), plot 3 (c) with two-dimensional slices and documented soil-pipe cross-section locations (black circles) from plot excavation with spline interpolated visualizations of continuous soil-pipes; decaying stump location indicated by white dots.

The ability to resolve soil pipes was also likely hindered by the reduction in data density surrounding the decaying tree stumps, where soil pipes with larger diameters primarily occurred within each plot (Leslie, 2012). The data coverage (a measure of certainty) decreases sharply with depth (Fig. 9), as is typical with

surface electrical resistivity surveys. However, the data coverage also decreases sharply in the area of the decaying stump due to the exclusion of electrodes placed in this area necessitated by the presence of the decaying stump. None of the stumps were completely consumed during the burning of the plots, and as a result, electrode placement continued to be hindered in this area for both pre- and post-burn measurements.

Summary and Conclusions

The goal of this study was to investigate the use of pseudo-three-dimensional ERT as a field method for characterizing soil-pipe spatial distribution and connectivity on a forested hillslope, with a specific interest in detecting the formation of soil pipes from tree decay and root combustion. Excavated and documented soil pipes with diameters larger than 1 cm showed a frequency increase with depth up to between 30 and 40 cm; with a median depth of 32 cm. This depth corresponded to the beginning of a dense BE horizon, which severely restricts root growth. This depth also corresponded with a high electrical-resistivity layer observed on all plots, which appears to be characteristic for the entire 2 ha field site. With most soil pipes containing <25% decaying root matter, there was a lack of resolvable contrast in electric properties between the highly resistive empty macropores and the compacted Bw/Be horizon within these hillslopes.

In conclusion, we argue that when mapping soil pipes or other cm-scale cavities, it is not practical to rely on homogeneity assumptions that are commonly used to detect anomalies characterized by higher resistivities compared to the surrounding soil. Even when using 25 cm grid spacing for electrodes—which is challenging experimentally—the resolution was limiting the detection of small changes in electrical resistivity, which was further hampered by the majority of large soil pipes occurring in low coverage regions. This means that it will be necessary to include either some vertical electrodes in the survey design, or enhance contrast by using tracers to improve detectability. Considering that the purpose of our study was to test if ERT could be used as a rapid and spatially exhaustive tool, we cannot recommend ERT for hillslope-scale characterizations of soil-pipe connectivity and distribution in the presence of resistive soil layers that are likely to have high resistivities.

Given that soil pipes often appear to form over restrictive soil and bedrock layers, the use of ERT surveys to detect macropore pathways occurring above or within these layers characterized by high electrical resistivity may be severely limited. ERT may be more successful in identifying soil pipes where the pathways are occurring within layers of low electrical resistivity, such as high clay-content horizons; however, only further research under field conditions can determine this.

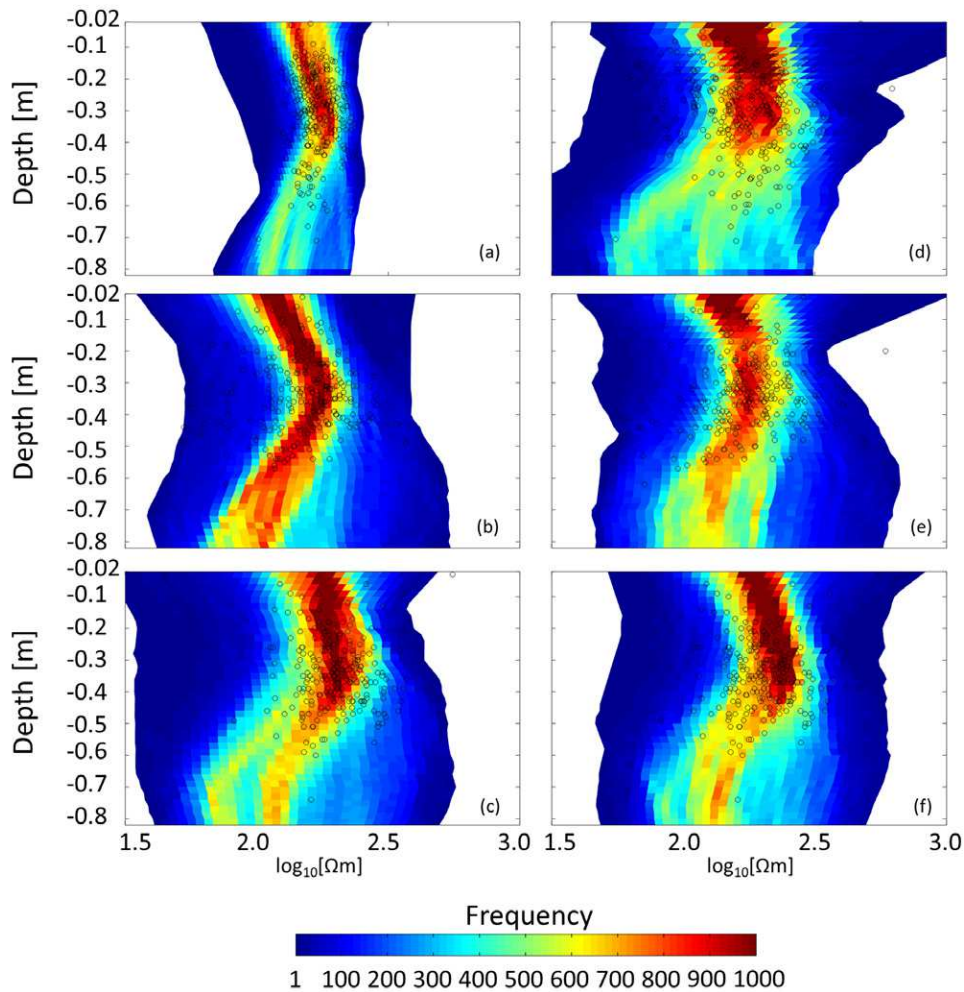


Fig. 8. Histograms of resistivity from pre-burn and post-burn plots plotted as a frequency with depth from -0.02 to -0.80 m in depth, and plotted with resistivity at soil-pipe cross-section locations, indicated by open black circles; (a): pre-burn plot 1, (b): pre-burn plot 2, (c): pre-burn plot 3, (d): post-burn plot 1, (e): post-burn plot 2, and (f): post-burn plot 3.

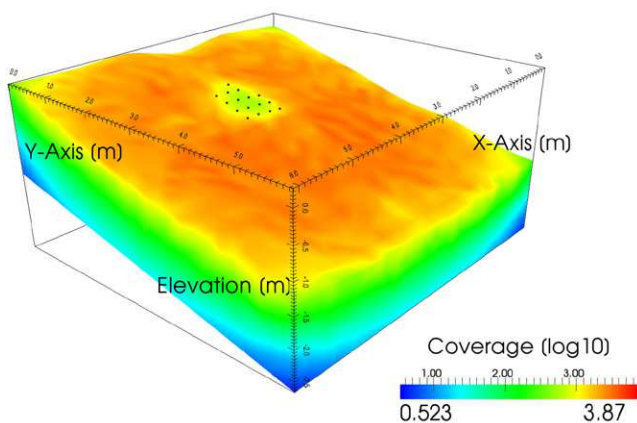


Fig. 9. Coverage distribution (sum of the absolute sensitivities) for the three-dimensional resistivity tomograms on example of plot 2 post-burn; positions where electrodes were omitted due to the presence of the decaying stump indicated by black dots.

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